

A Hybrid, Density-Based Clustering Approach for Spatial Settlement Graphs

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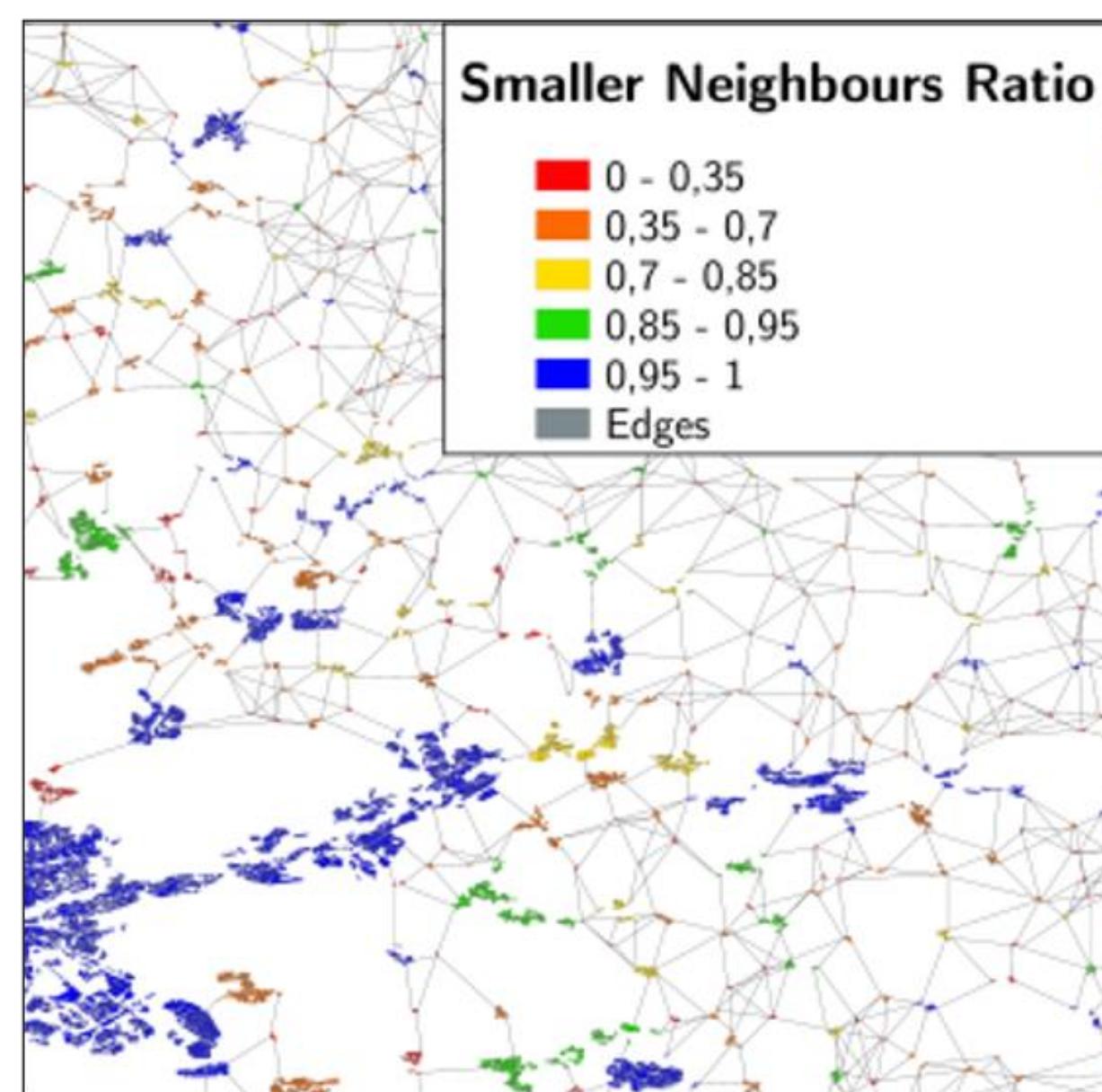
How realistic is this clustering of urban areas?

Problem

- Settlement chains and weak compactness in current clustering
- Existing graph-based methods too sensitive to distant edges
- Need for semantic-aware spatial grouping

Research questions

- Can DBSCAN/DBSC improve clustering quality?
- What metrics are best for evaluating results?
- How to integrate this with FISHNET existing framework?



Methodology

- WSF data preprocessing graph construction
- Clustering: DBSCAN/DBSC
- Qualitative metrics: connectivity, compactness

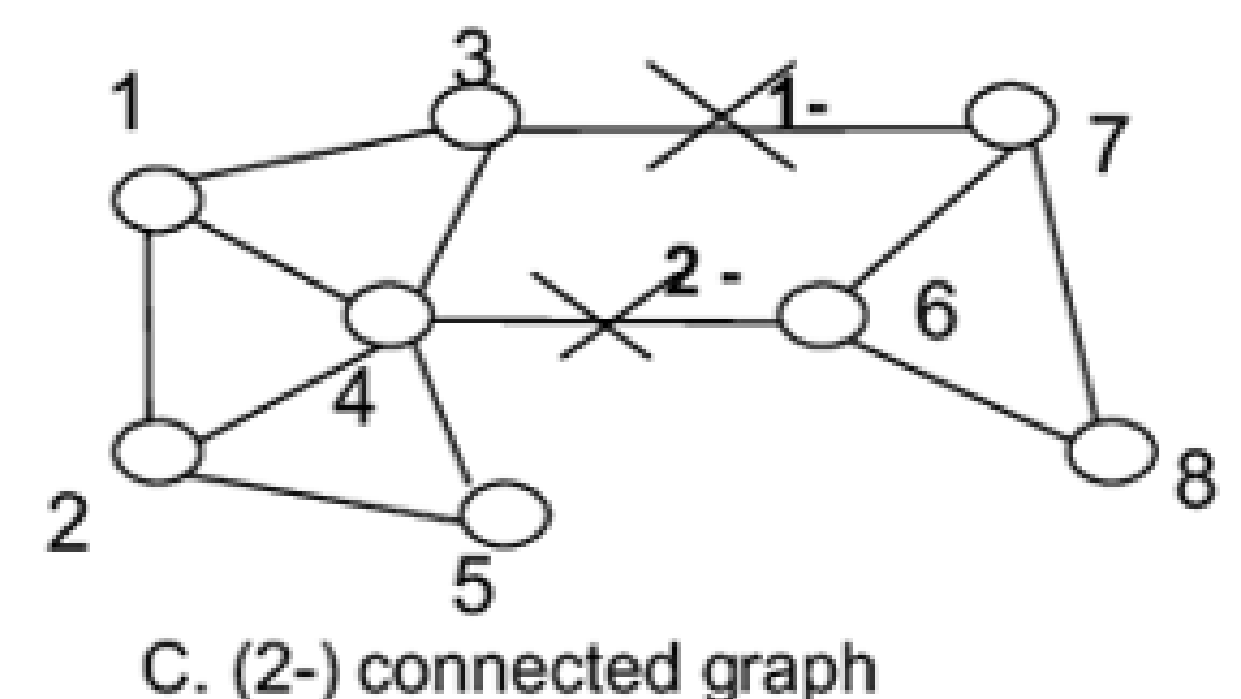
Visualisation

Clusters are visualized using color-coded spatial maps in QGIS to support expert analysis and detect spatial anomalies.

Evaluation techniques



- **Compactness (CI):** Measures how tightly the clustered polygons are grouped.
- **Connectivity (κ):** Evaluates how well-connected the elements inside a cluster are within the graph structure.
- **Runtime Performance:** Measures efficiency in computation, relevant for large spatial datasets.



Expected Results

- The DBSC-enhanced workflow is expected to **reduce settlement chaining**, producing more compact and functionally coherent urban clusters. Clusters will better reflect **real-world spatial relationships and population distribution**.
- The method aims to produce **clearer visual delineations** of settlement boundaries, aiding planners and researchers.
- If implemented as planned, this could support **more accurate infrastructure planning** and **urban growth modeling** in heterogeneous environments.

